

STONY RAPIDS, SK, Canada

WMO: 711320

Lat: 59.250N

Lon: 105.840W

Elev: 244

StdP: 98.43

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.1	-38.4	-44.1	0.1	-40.6	-42.4	0.1	-38.2	7.6	-18.0	7.0	-19.9	0.6	100	0.712

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	27.8	16.5	26.1	16.0	24.1	15.3	18.2	24.6	17.2	23.2	16.3	22.0	3.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	11.8	19.6	15.1	11.0	19.1	14.0	10.2	18.2	52.6	24.4	49.5	23.4	46.7	22.0	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.4	6.4	5.7	DB	-43.2	31.2	2.4	1.8	-45.0	32.4	-46.4	33.5	-47.8	34.5	-49.5	35.8	(4)
(5)			WB	-42.9	20.2	2.0	1.2	-44.4	21.1	-45.6	21.8	-46.7	22.5	-48.2	23.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.9	-24.0	-21.0	-13.7	-3.3	5.9	13.4	16.2	14.3	8.3	0.1	-11.6	-20.4	(6)
(7)		DBStd	15.56	8.68	7.91	8.64	7.44	5.22	3.92	3.03	3.37	4.20	4.80	7.43	8.70	(7)
(8)		HDD10.0	5200	1053	868	736	398	147	14	1	5	81	308	647	943	(8)
(9)		HDD18.3	7772	1311	1101	994	648	386	154	79	133	302	566	897	1201	(9)
(10)		CDD10.0	495	0	0	0	1	20	114	193	137	29	1	0	0	(10)
(11)		CDD18.3	25	0	0	0	0	1	5	13	6	0	0	0	0	(11)
(12)		CDH23.3	563	0	0	0	0	23	143	267	121	9	0	0	0	(12)
(13)		CDH26.7	109	0	0	0	0	4	23	56	25	1	0	0	0	(13)
(14)	Wind	WSAvg	2.4	2.1	2.2	2.5	2.7	2.9	2.6	2.3	2.2	2.4	2.3	2.1	1.9	(14)
(15)	Precipitation	PrecAvg	400	15	15	17	15	25	39	61	73	55	33	23	13	(15)
(16)		PrecMax	623	42	92	42	36	99	84	154	130	190	68	70	31	(16)
(17)		PrecMin	258	1	0	0	0	3	1	31	18	17	10	3	1	(17)
(18)		PrecStd	87	11	17	11	10	20	23	26	29	32	14	16	9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.2	0.0	8.4	17.3	26.1	29.1	30.5	29.8	24.5	15.9	4.3	-0.9	(19)
(20)			MCWB	-1.4	-1.0	3.1	8.5	13.8	16.4	18.1	18.5	15.9	10.3	2.0	-1.7	(20)
(21)		2%	DB	-5.5	-3.6	5.0	13.1	22.3	26.9	28.2	26.4	20.7	11.5	1.4	-4.0	(21)
(22)			MCWB	-6.0	-4.4	1.3	6.3	11.9	15.2	17.0	16.9	13.9	7.7	0.5	-4.5	(22)
(23)		5%	DB	-8.2	-6.3	1.8	10.3	19.2	24.9	26.4	24.2	17.8	8.8	0.1	-6.1	(23)
(24)			MCWB	-8.7	-7.0	-0.6	4.7	10.5	14.7	16.5	16.2	12.2	6.1	-0.8	-6.6	(24)
(25)		10%	DB	-11.5	-9.1	-1.1	7.5	16.1	22.7	24.4	21.9	15.4	6.6	-1.7	-8.3	(25)
(26)			MCWB	-11.9	-9.8	-2.8	3.1	9.0	13.8	15.8	15.1	11.4	4.8	-2.5	-8.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.3	-0.9	3.5	8.9	14.7	18.7	19.8	19.6	16.5	10.9	2.3	-1.6	(27)
(28)			MCDWB	-1.0	-0.5	8.1	16.3	23.7	24.4	26.6	26.8	23.4	14.4	3.6	-1.0	(28)
(29)		2%	WB	-5.9	-4.4	1.4	6.6	12.7	17.0	18.5	17.9	14.5	7.9	0.6	-4.5	(29)
(30)			MCDWB	-5.4	-3.7	4.5	12.4	20.4	23.6	25.0	24.4	18.9	10.7	1.3	-4.1	(30)
(31)		5%	WB	-8.7	-7.0	-0.4	5.0	11.0	15.6	17.5	16.9	13.1	6.4	-0.8	-6.6	(31)
(32)			MCDWB	-8.3	-6.3	1.7	9.6	18.3	22.9	23.6	22.3	16.6	8.5	0.1	-6.1	(32)
(33)		10%	WB	-11.9	-9.7	-2.8	3.4	9.4	14.5	16.7	15.8	11.9	4.9	-2.4	-8.7	(33)
(34)			MCDWB	-11.5	-9.1	-1.1	7.2	15.4	21.3	22.4	20.4	15.0	6.5	-1.7	-8.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.2	11.4	13.8	13.1	12.8	13.2	12.9	11.6	9.9	6.8	7.6	8.6	(35)
(36)			MCDBR	10.0	11.0	14.6	15.5	18.8	17.9	17.4	16.7	15.3	10.9	7.5	9.0	(36)
(37)			MCWBR	9.8	10.5	11.2	9.6	9.8	8.2	7.8	8.1	8.7	7.3	6.8	8.8	(37)
(38)		5% WB	MCDBR	10.0	10.9	14.2	14.1	17.1	15.1	14.0	14.2	13.0	9.8	7.3	8.8	(38)
(39)			MCWBR	9.8	10.5	11.1	8.8	9.3	7.4	6.9	7.4	7.8	6.8	6.7	8.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.214	0.254	0.259	0.322	0.330	0.364	0.403	0.358	0.310	0.286	0.230	0.182	(40)	
(41)		taud	2.166	2.230	2.303	2.226	2.436	2.396	2.317	2.434	2.588	2.520	2.216	2.083	(41)	
(42)		Ebn at Noon	631	778	896	880	901	870	821	838	839	750	594	539	(42)	
(43)		Edh at Noon	42	69	88	115	101	107	114	94	69	54	40	30	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.50	1.42	3.11	4.92	5.77	5.94	5.31	4.12	2.63	1.22	0.43	0.28	(44)	
(45)		RadStd	0.04	0.11	0.19	0.34	0.58	0.52	0.30	0.27	0.29	0.13	0.05	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page